

Case Study: LinkedIn AI Post Studio

Role: Solo Scrum practitioner (PO / SM / Developer)

Project: Local Python Web App for AI Content Automation (2025–2026)

Release covered: On-Demand Publishing rebuild → Fact Check → Versioned API → SEO / AEO

Agile Governance & Product Ownership (The "What")

- **Vision to Backlog Transformation:** Acted as the sole Product Owner to translate high-level personal branding goals into a structured backlog of 35+ category features and 20+ user stories across 7 content domains (AI, cybersecurity, cloud, GRC, pharma, and more).
- **Value-Based Prioritization:** Decomposed a complex multi-phase workflow (trend fetch → draft → review → publish) into independently shippable increments, ensuring a functional "Minimum Viable Product" by the end of Sprint 1 and protecting a predictable delivery cadence thereafter.
- **Decisive Scope Reduction (Descoping as a Feature):** Rather than defending a bloated backlog, exercised hard PO discipline by formally removing the scheduler and analytics epics once the underlying OAuth constraint made them low-value and high-maintenance. Re-anchored the product around a leaner "on-demand publishing" vision — cutting technical debt, sharpening the value proposition, and increasing sprint velocity on the features users actually reached for.
- **Strategic Scope Guardrails:** Maintained a documented out-of-scope register (multi-account support, hosted SaaS deployment, autonomous multi-account posting) to prevent scope creep and keep every sprint anchored to the single-user, locally-run product vision.

Scrum Mastery & Process Facilitation (The "Process")

- **End-to-End Sprint Management:** Facilitated the full agile lifecycle across multiple releases, maintaining strict iterative discipline from initial discovery through the on-demand rebuild, the Fact Check release, the versioned-API migration, and the current SEO / AEO release.
- **Impediment Removal & Risk Mitigation:** Proactively identified LinkedIn's `r_member_social` OAuth restriction as a hard blocker for the analytics/insights features. Instead of stalling the sprint on an external dependency outside the team's control, pivoted transparently — descoping analytics, documenting the constraint, and redirecting that capacity toward higher-value content-quality features (Fact Check and SEO / AEO).
- **Retrospective-Driven Improvement:** Ran personal retrospectives that converted friction into concrete backlog items — including a "regenerate-without-refetch" flow to cut draft-iteration cycle time, and the simplification decision that retired unused scheduling complexity.
- **Protecting the Increment (Zero-Regression Delivery):** Delivered the SEO / AEO release as a strictly additive, off-by-default increment. Existing draft and publishing behavior remained byte-for-byte unchanged for users who left the toggle off, honoring the Scrum value of a "Done," non-breaking increment every sprint and eliminating regression risk on shipped functionality.

Technical Execution & Architecture (The "How")

- **Cross-System Integration:** Executed and maintained integrations across 5 external systems (LinkedIn REST API, Abacus.AI LLM, DuckDuckGo search, 50+ RSS feeds, and Google Trends), establishing robust API contracts, timeouts, and graceful fallback strategies so a single upstream failure never breaks the workflow.

- **Platform Deprecation Management:** Treated LinkedIn's API deprecation timeline as a first-class technical risk and migrated the entire publishing pipeline to the versioned LinkedIn REST API (LinkedIn-Version: 202507, /rest/posts and /rest/images), future-proofing the app's core "go-live" capability against legacy-endpoint sunset.
- **Quality Engineering — Content Trust:** Engineered a Fact Check capability that extracts 3–5 verifiable claims from a draft, searches live web sources for each, and scores the post against real evidence (verified / uncertain / likely-false) — grounding AI output in current reality rather than model training data.
- **Discoverability Engineering — SEO / AEO:** Shipped an optional SEO / Answer-Engine-Optimization layer that tunes drafts and hashtags for search and AI answer engines (ChatGPT, Perplexity, Google AI Overviews), plus an AEO Score that rates any post 0–100 across keyword presence, quotability, claim clarity, and structure with concrete improvement suggestions.
- **Technical Transparency:** Maintained a "living" technical specification — including .env guides, API endpoint tables, LLM token budgets, and an architecture overview in the README — to minimize technical debt and keep the codebase maintainable by a solo practitioner.

Operational Resilience

- **Dependency Management:** Managed the recurring 60-day LinkedIn OAuth token expiry cycle as a high-priority operational risk; built a debug validation endpoint to catch expired credentials early and mitigate "go-live" publishing failures.
- **Data Freshness Engineering:** Responded to duplicate-content feedback by implementing a 30-day TTL "seen-articles" cache, eliminating repeat suggestions and guaranteeing users see fresh source material on every fetch.